

I have to check but if that is the error wat should I do to fix it

Simple fix — move `v_addr := ahbmo.haddr` inside the NONSEQ case only:

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Current (Wrong):

```
vhdl
process(rst, clk) is
begin
    v_addr := ahbmo.haddr;  -- ← runs EVERY clock edge, overwrites v_addr!

    if rst = '0' then
        ...
    else
        if rising_edge(clk) then
            case v_trans is
                when NONSEQ =>
                    count := 0;
                    ...
                when SEQ =>
                    v_addr := v_addr + 4;  -- too late, already overwritten above
```

Fixed:

```
vhdl
process(rst, clk) is
begin
    -- DO NOT capture v_addr here anymore

    if rst = '0' then
        ...
    else
        if rising_edge(clk) then
            case v_trans is
                when NONSEQ =>
                    v_addr := ahbmo.haddr;  -- ← capture ONLY here
                    count := 0;
                    ...
                when SEQ =>
                    -- v_addr is now purely incremented from previous value
                    count := count + 1;
                    v_addr := std_logic_vector(
                        unsigned(v_addr) + to_unsigned(4, v_addr'length));
```

Why This Works

Clock 1 (NONSEQ): `v_addr := haddr = 0x100` → `inter_addr=0x100` ✓
Clock 2 (SEQ1): `v_addr = 0x100+4 = 0x104` → `inter_addr=0x104` ✓
Clock 3 (SEQ2): `v_addr = 0x104+4 = 0x108` → `inter_addr=0x108` ✓
Clock 4 (SEQ3): `count=3, flag=1, no update` → `inter_addr=0x10C` ✓

No double increment, no skip.

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